1. YIWEI JIA ON USE-CASE 2: EYE-TRACKER

Use case 2: Eye-Tracker

Questions:

- Channel: Human eye-movements via infra-red camera
- Environment: Dark room, possible head movements
  - Noise: Blinks, Eye-movements, Sampling failure (device error)
  - How can I filter this noise: Erase data points affected by artefacts
- Dimensionality of the signal: 1D and 2D (why?)
- What is the sampling rate? Infra-red camera specifics
- Sample size: 4

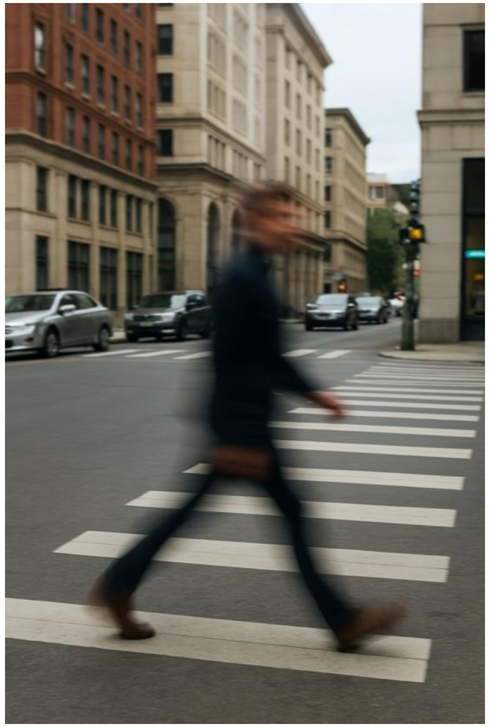
Which process am I investigating?

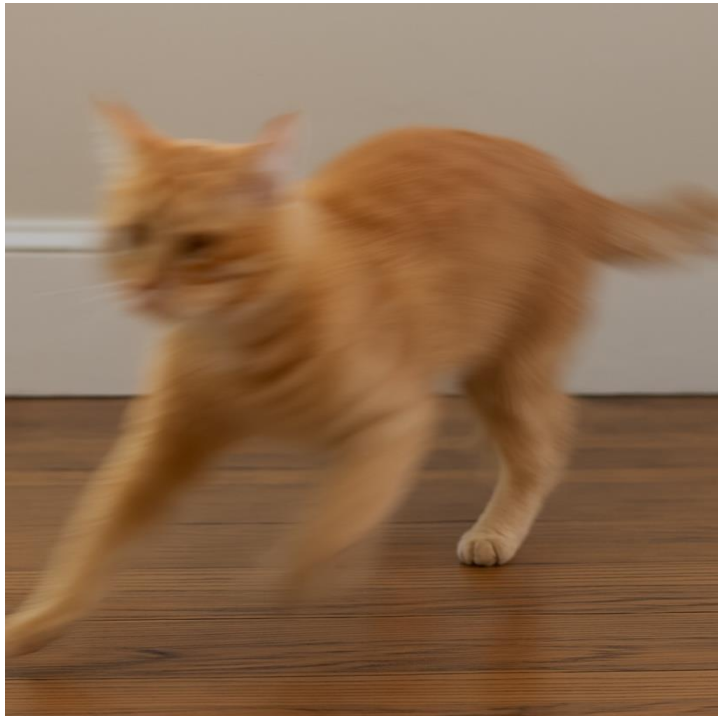
- Gaze stability (fixation ability of the subjects), Deviation from center

Use case 2: Eye-Tracker

Context of eye tracking data processing:

- **Motion artifact** : slow exposure + moving subjects = **blur**.





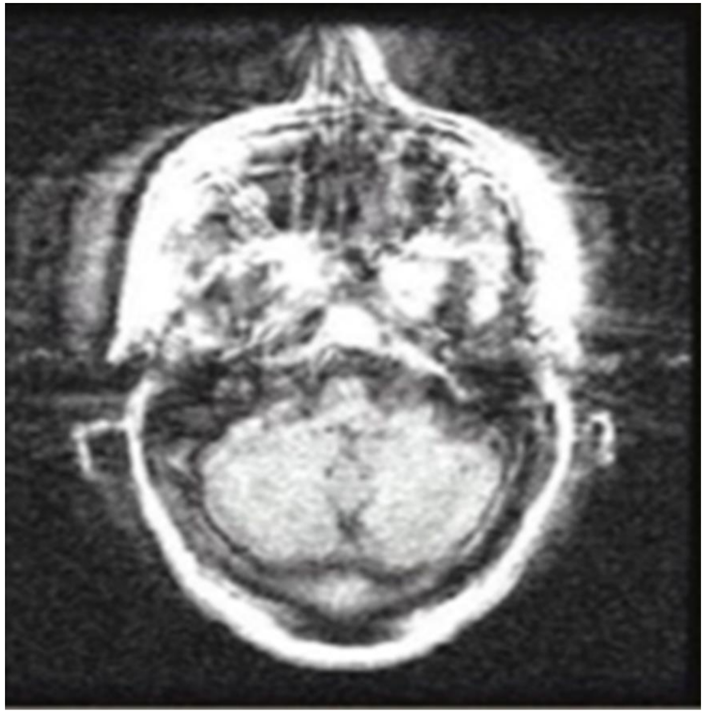
[1] https://www.pearl-technology.ch/en/blog/motion-artifacts_prevalence-sideeffects-winningstrategies

Use case 2: Eye-Tracker

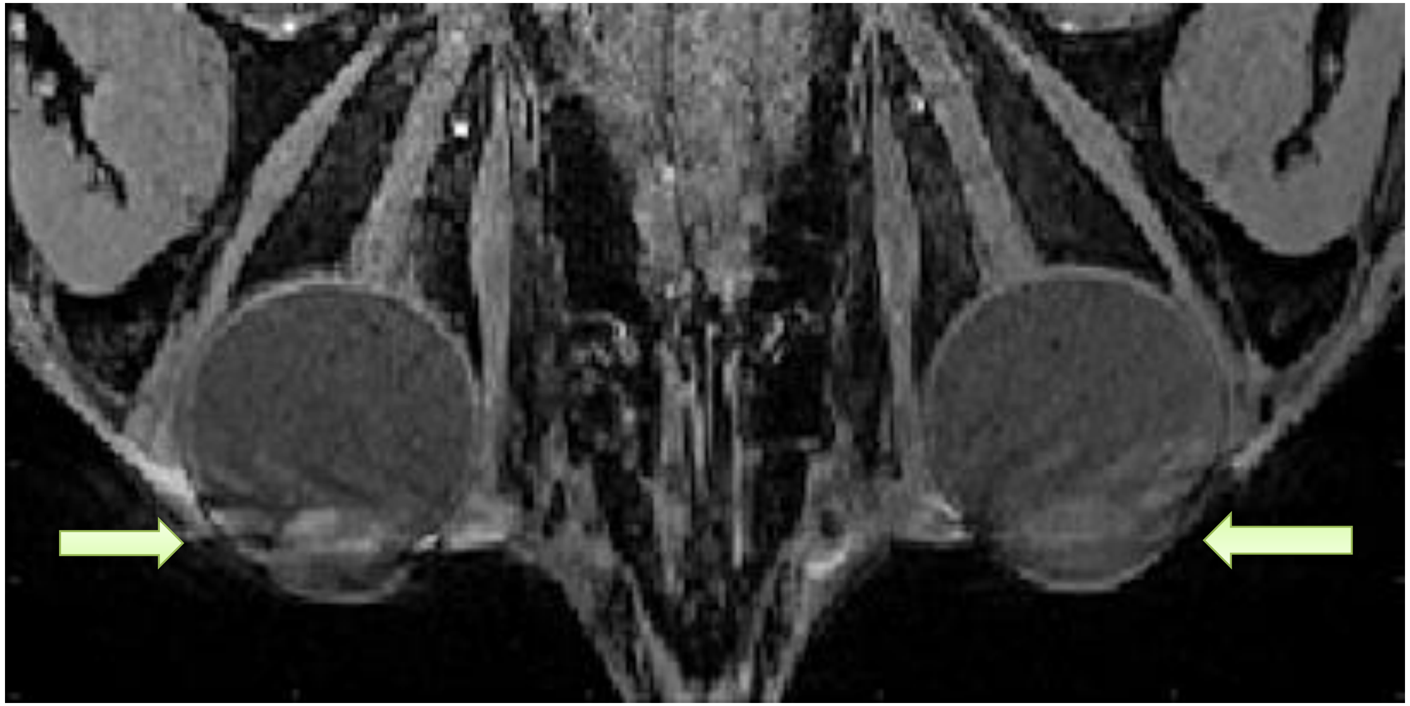
Context of eye tracking data processing: Motion artifact

Long acquisition time of MRI +

Head movement



Rapid eye movement



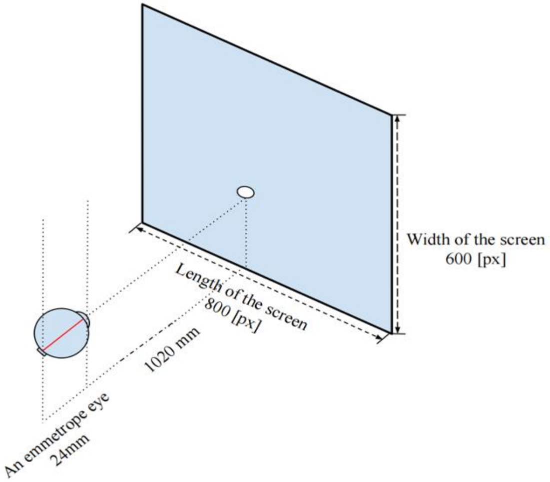
[1] https://www.pearl-technology.ch/en/blog/motion-artifacts_prevalence-sideeffects-winningstrategies

Use case 2: Eye-Tracker

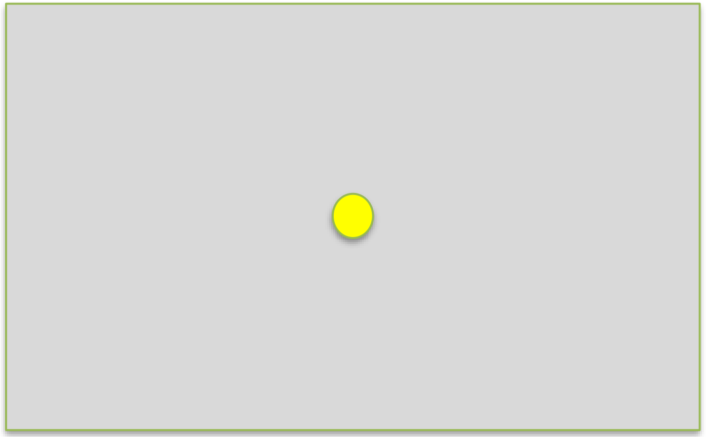
Context of eye tracking data processing:

- **Motion artifact** :
  Long acquisition time of MRI + **Rapid** eye movement

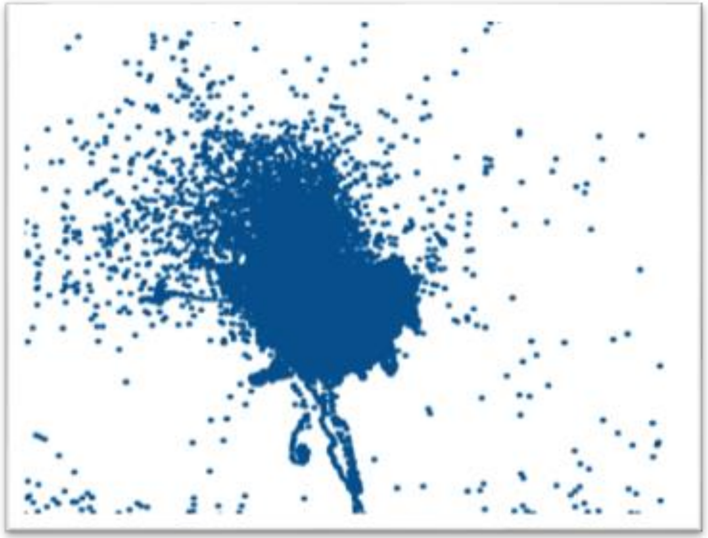
- **Method: Track the eye movement in real-time**
  - Records: gaze position
  - Types of eye movement: Blinks, Saccades, Tremor
  - Goal: eliminate MRI data affected by motion



A dot at the center of screen



Gaze positions (one MRI session)



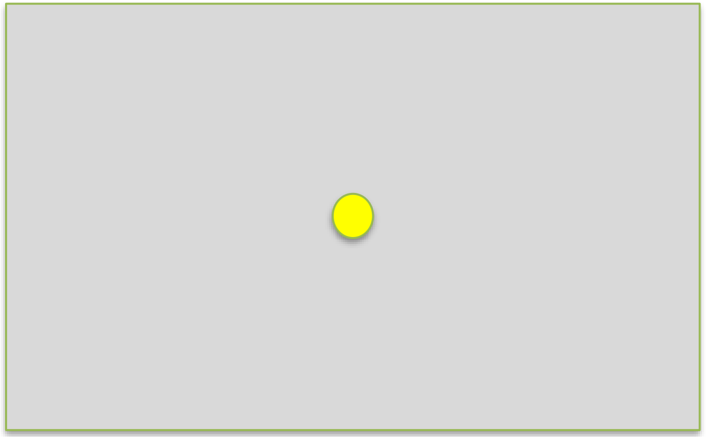
Use case 2: Eye-Tracker

Context of eye tracking data processing:

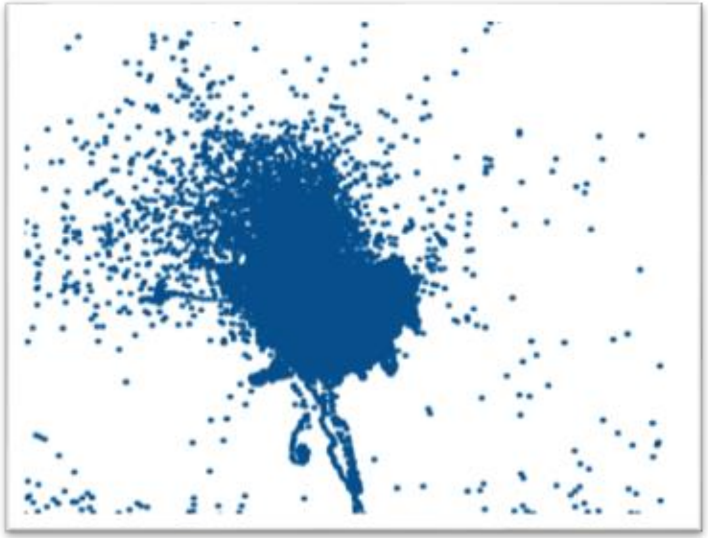
- **Motion artifact** :
  Long acquisition time of MRI + **Rapid** eye movement

- **Method: Track the eye movement in real-time**
  - Records: gaze position
  - Types of eye movement: Blinks, Saccades, Tremor
  - Goal: eliminate MRI data affected by motion

A dot at the center of screen



Gaze positions (one MRI session)



Use case 2: Eye-Tracker

Context of eye tracking data processing:

- **Motion artifact** :
  Long acquisition time of MRI + **Rapid** eye movement

- **Method: Track the eye movement in real-time**
  - Records: gaze position
  - Types of eye movement: Blinks, Saccades, Tremor
  - Goal: eliminate data affected by motion

- **What we do with ET data**
  - Filter Noise: device error (eg. extremely large values)
  - Eliminate data points affected by motions.
  - Investigate Eye Movement Features
  - Understand the characteristics, quality control

Use case 2: Eye-Tracker

Part 1: Know your data at the first glance
(parsing metadata/ visualization)

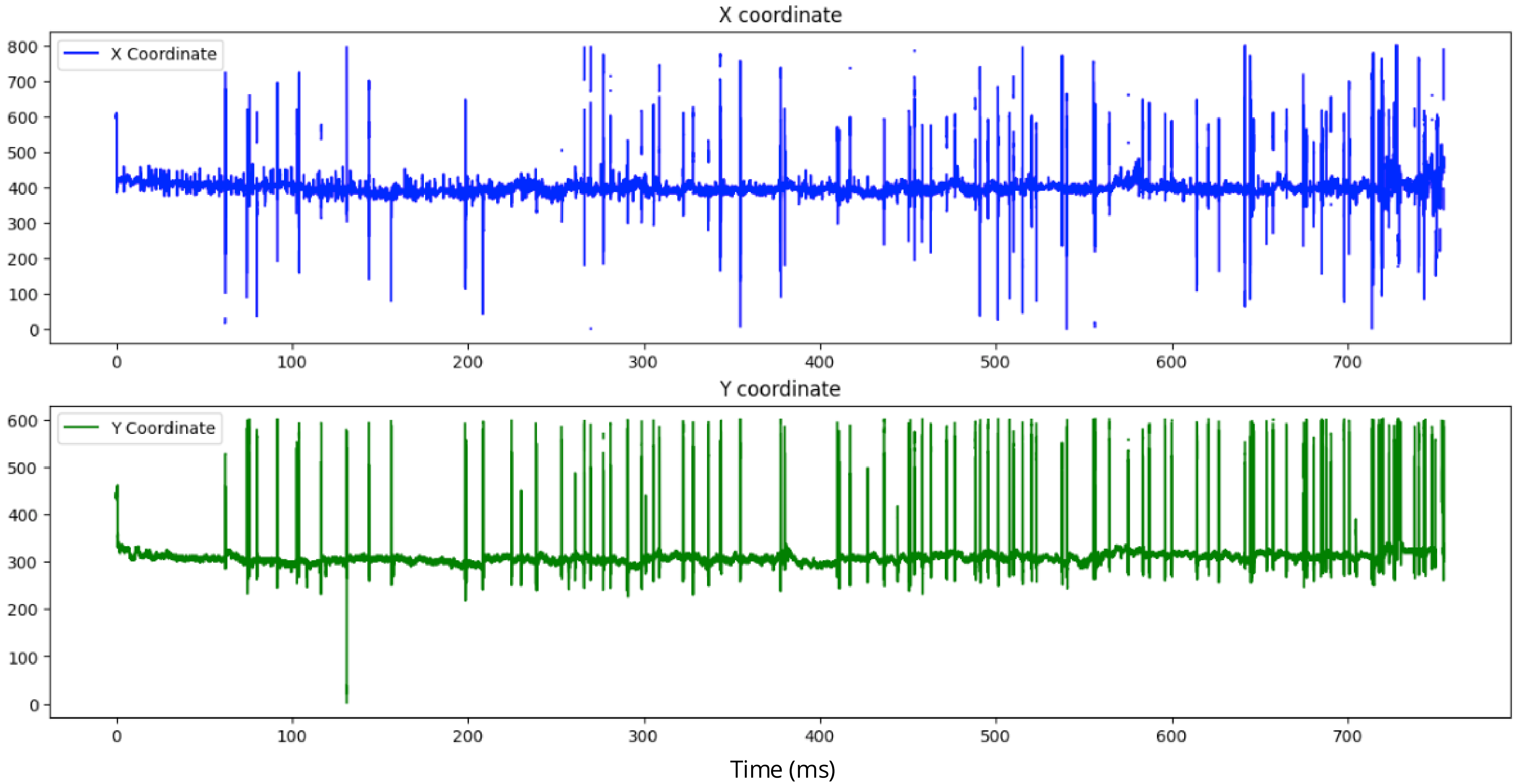
- Load the data and parse the metadata
- Plot each dimension of gaze trajectories over time
- Overlay different event types (saccades, blinks, fixations) using different colors or markers.

What is the correlation between the events and gaze trajectories?

- Do **fixations** correspond to relatively **flat** or stable gaze lines?
- Do **saccades** appear as **sharp jumps**?
- Are there any periods where the **gaze suddenly drops out**, possibly indicating **a blink**?

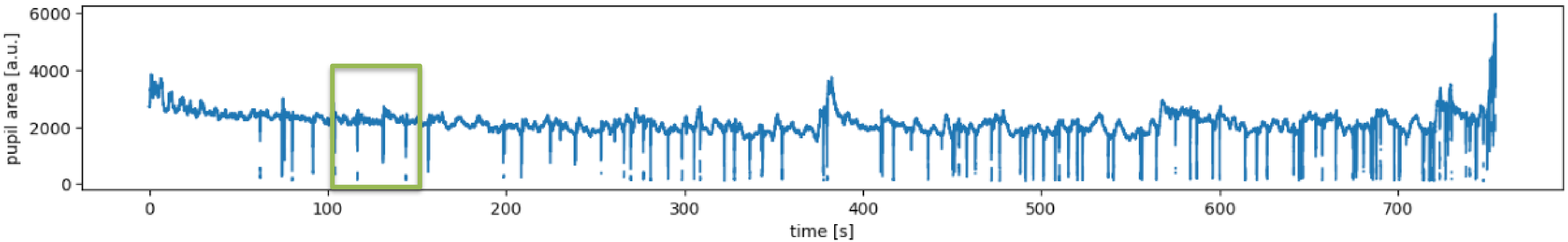
Use case 2: Eye-Tracker

Take Subject 1 as example:
Check the gaze trajectory in two dimensions along time axis

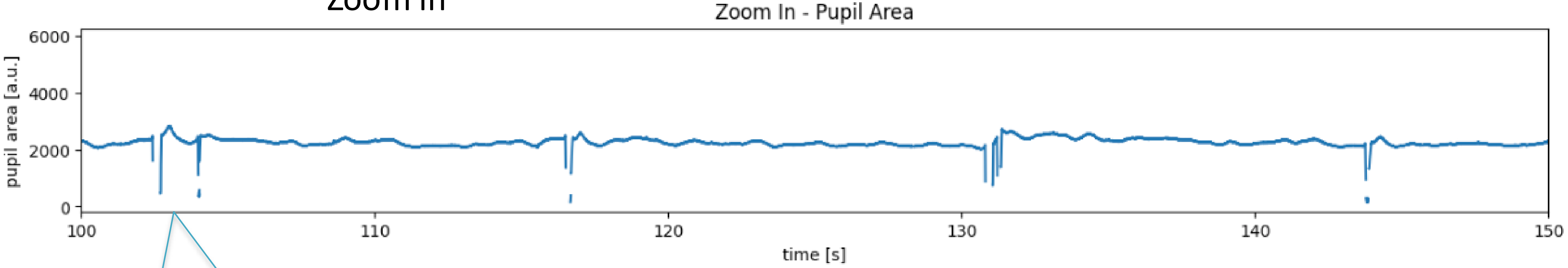


Use case 2: Eye-Tracker

Check the **pupil area** in two dimensions along time axis



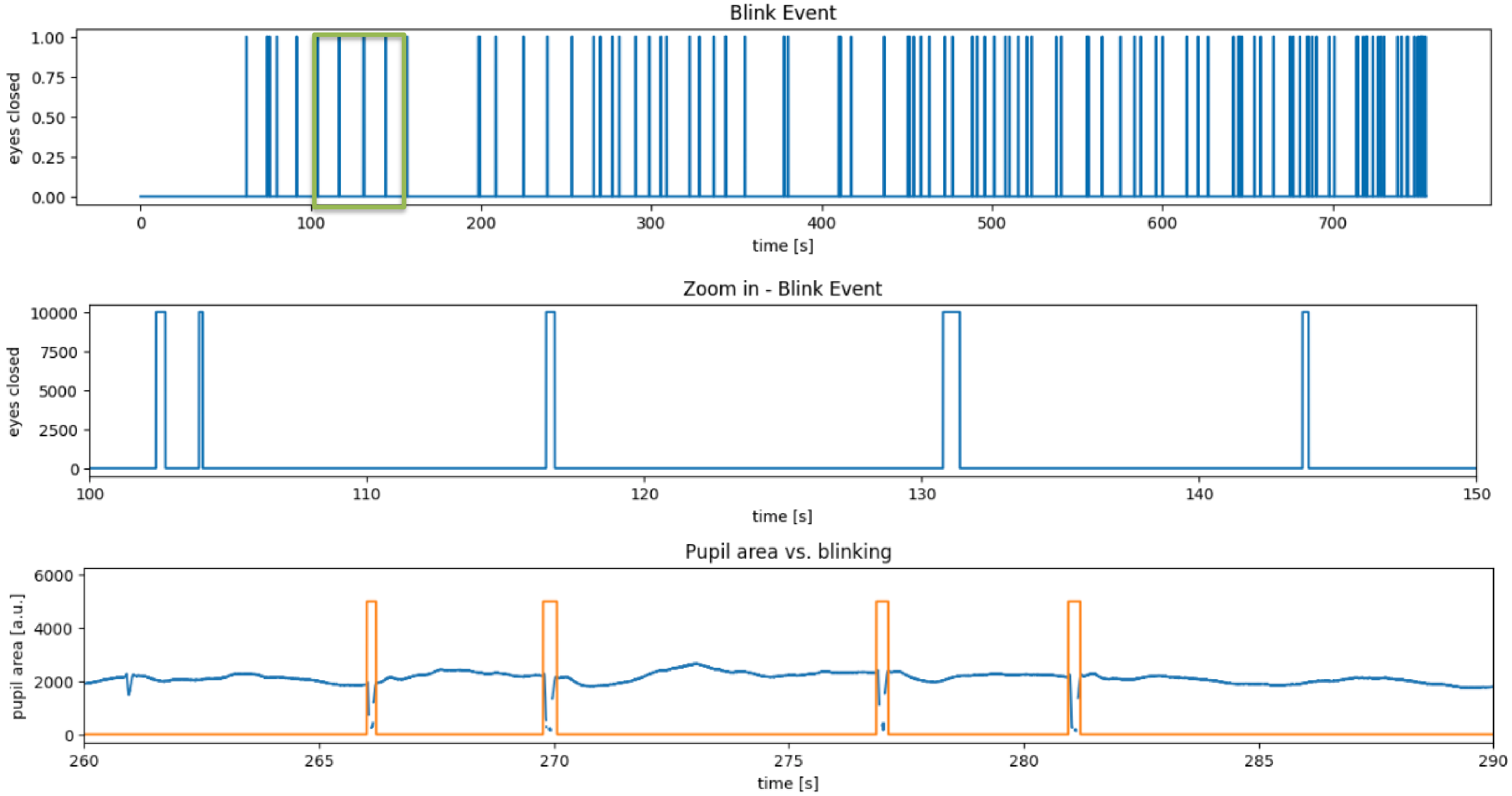
Zoom in



discontinuous
Profiles ?

Use case 2: Eye-Tracker

Overlap the **blink event** with pupil area

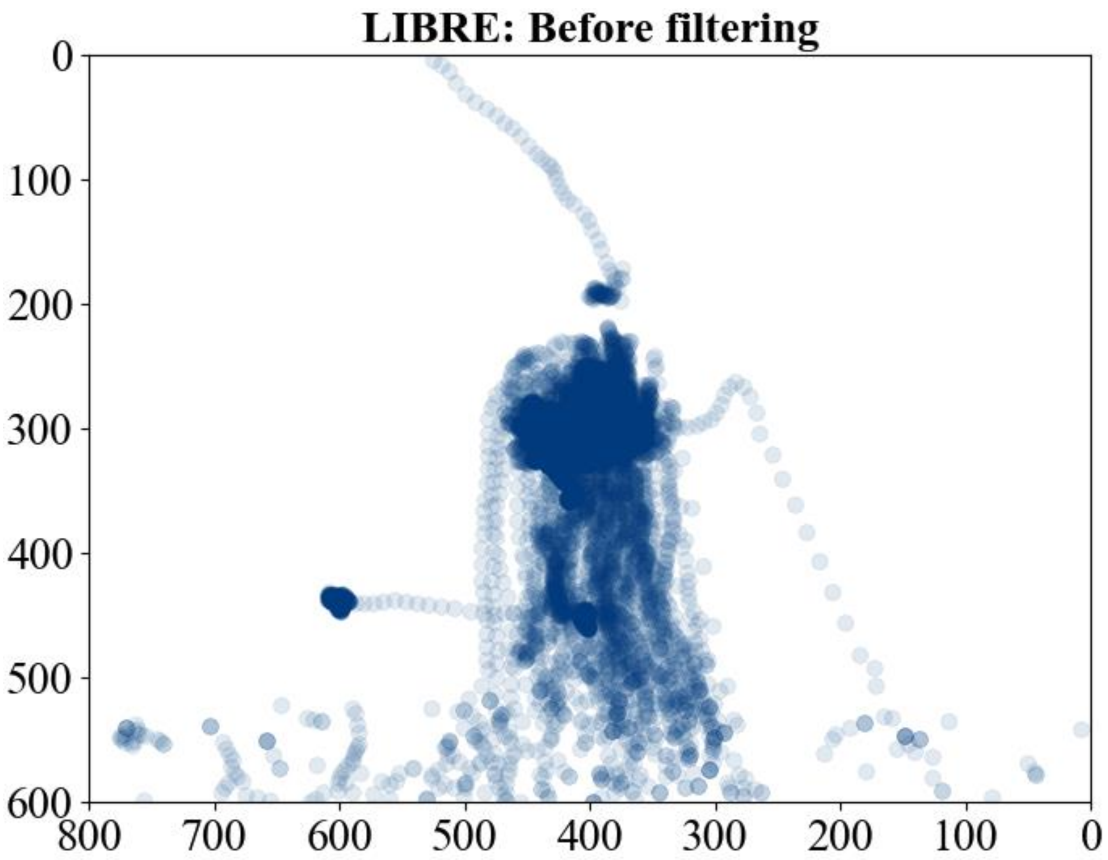


Use case 2: Eye-Tracker

Check the x coordinate of gaze trajectory with **saccade**

Use case 2: Eye-Tracker

Visualize the gaze point in 2D



Points deviated from the center of the screen

Use case 2: Eye-Tracker

Part 2: Feature analysis

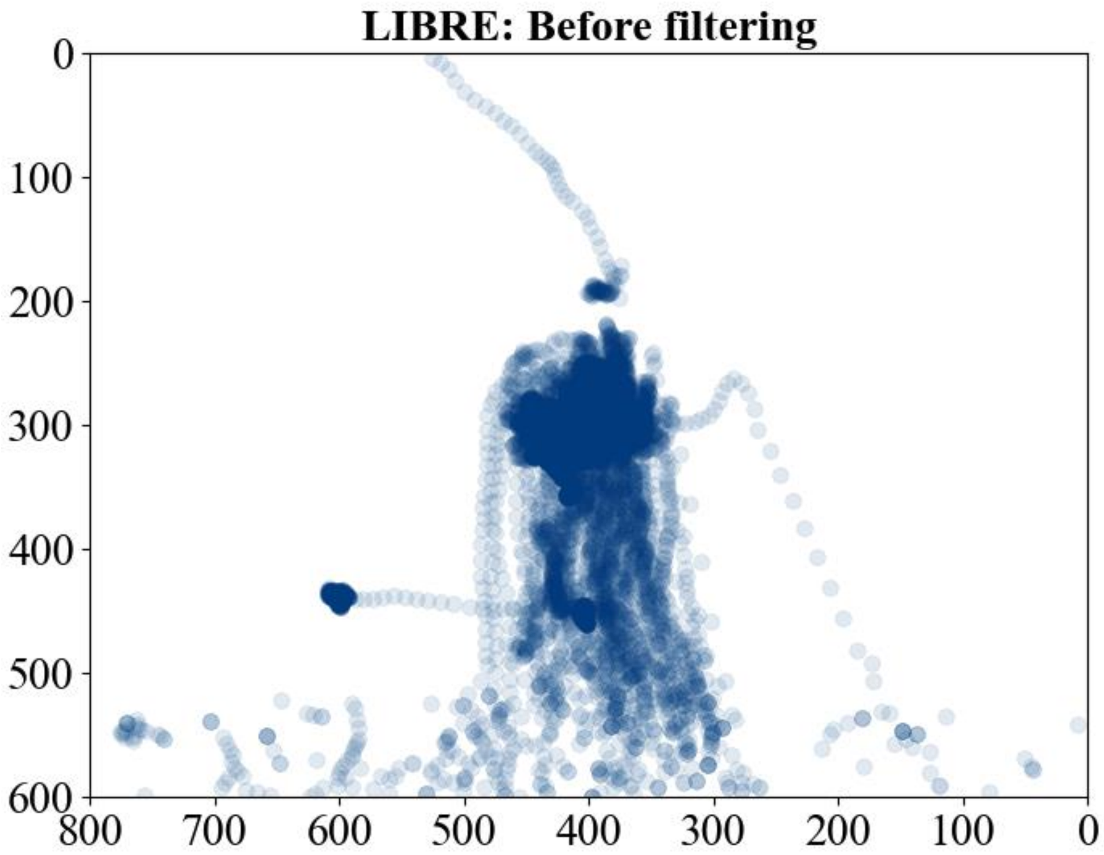
- **Distribution** of gaze points along each direction.
- Determine the **center** of gaze points and the **deviation**.
- Calculate **statistics** (mean, median, standard deviation) for each subject
  - Saccade duration, blink duration, and fixation duration.

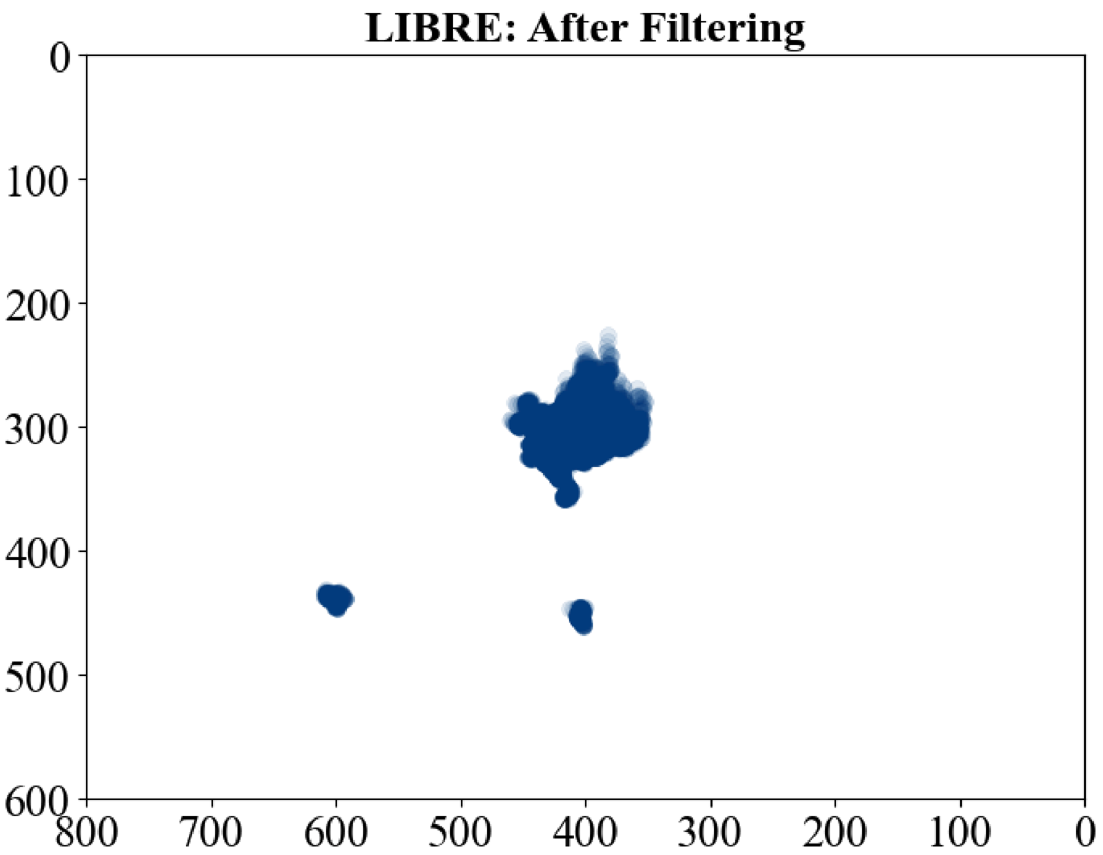
  → Solid set of features per subject

- Quality analysis
  - Do the features make sense compared to expected behavior?
  - Do some subjects need to be excluded from further analysis due to poor data quality?

Use case 2: Eye-Tracker

Filter out the gaze points affected by **blinks and saccades**
Preserve data during the **fixation**





Use case 2: Eye-Tracker

Calculate the statistics of the gaze points along each dimension

Statistic	x-Dimension	y-Dimension
Mean	397.07	305.27
Median	395.5	305.30
Standard Deviation	13.96	9.86
Minimum	353.40	225.80
Maximum	609.80	460.70
25th Percentile (Q1)	388.30	301.00
75th Percentile (Q3)	405.30	309.00

Recall: Cours4

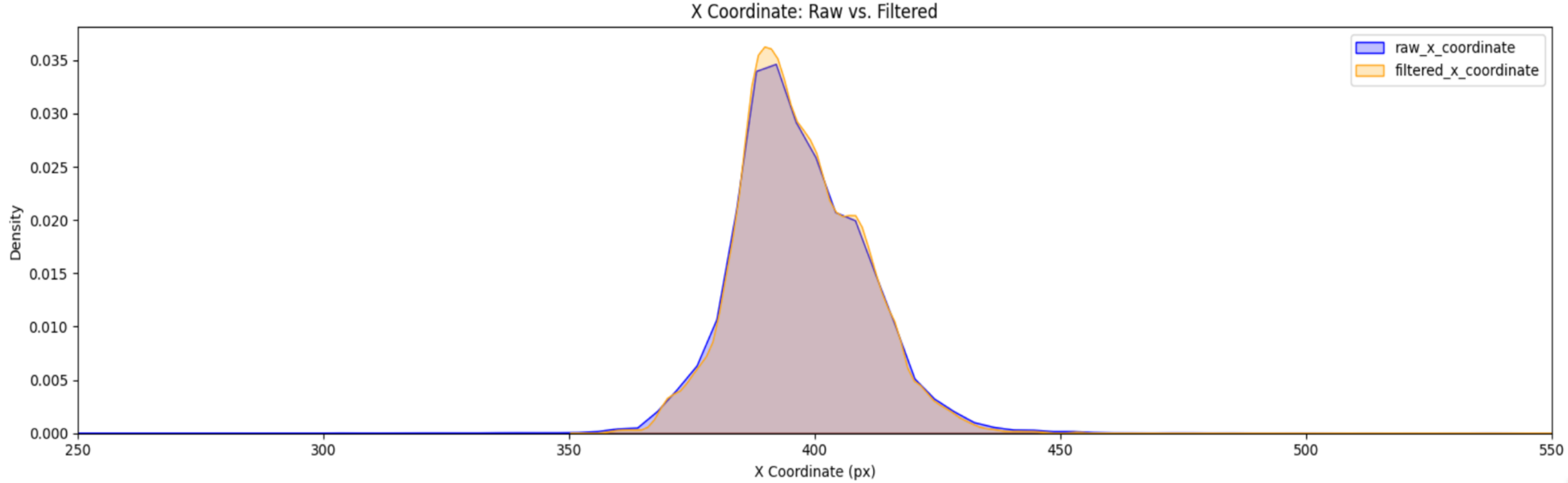
Descriptive statistics of a time series

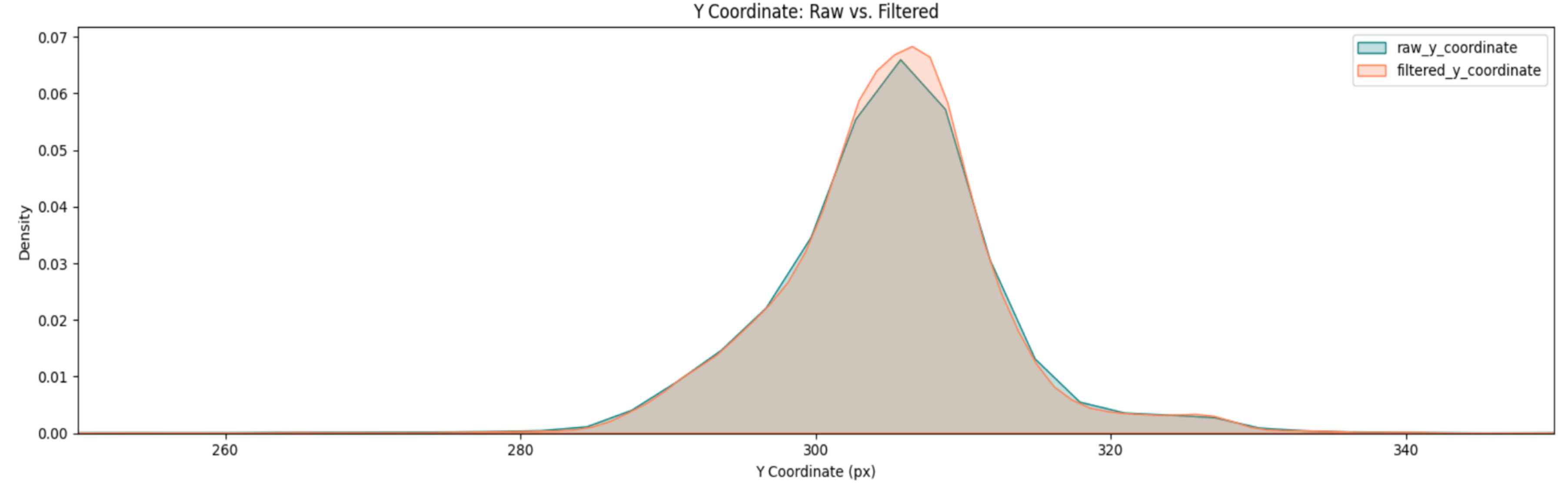
Signal characterization:

1. **Max, Min, Mean** - Central value of the signal.
2. **Median**: Middle value when sorted.
3. **Variance & Standard Deviation**: Measure of signal spread (and fluctuation).
4. **Distribution** analysis: Shows patterns like normal or skewed distribution.

Use case 2: Eye-Tracker

Investigate the density of gaze points along each dimension (before filtering vs. after filtering)





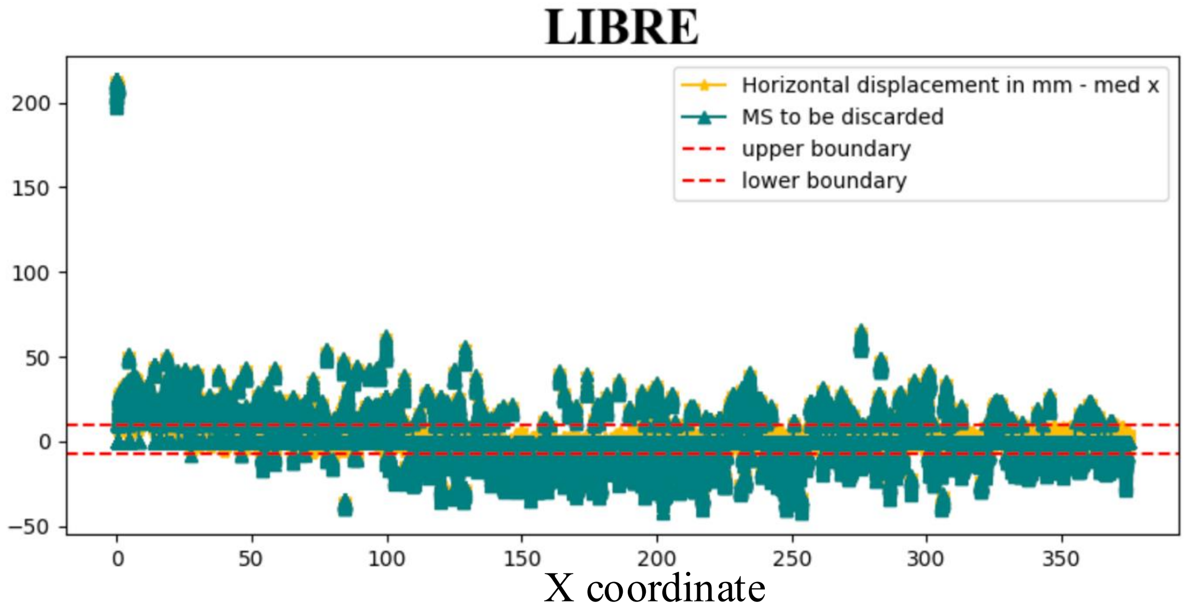
How has the distribution changed after filtering?

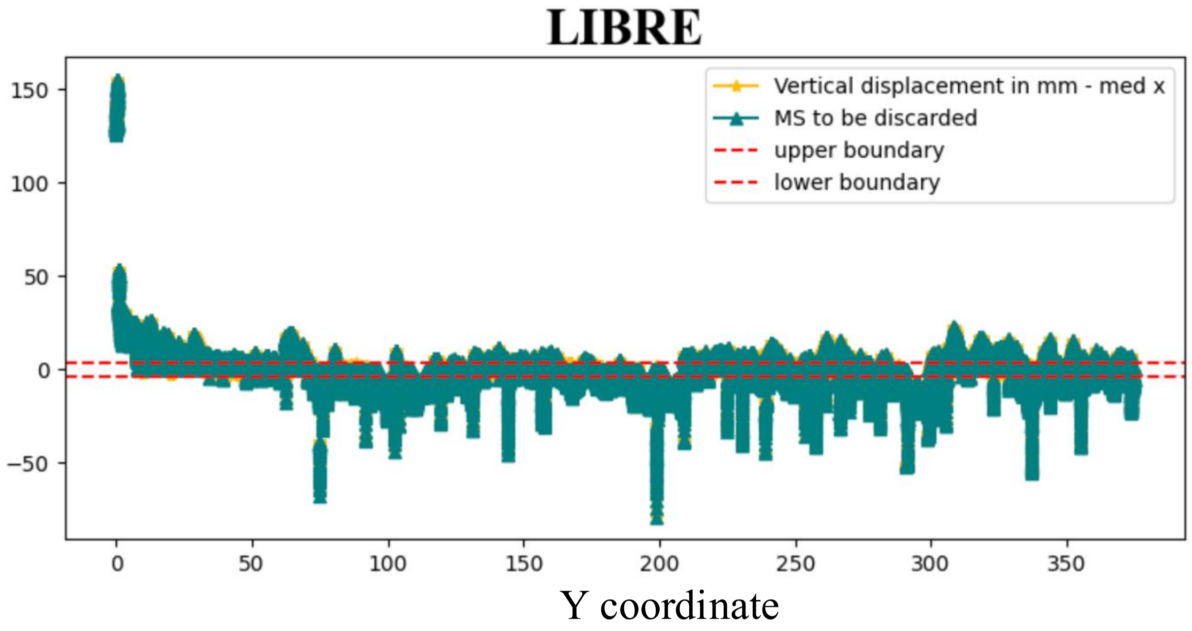
Use case 2: Eye-Tracker

Check Quartiles: Values that divide a data set into four equal parts after it has been sorted.

- **Q1 (25th percentile)**: The value that separates the lowest 25% of the data. 25% of the data points are less than Q1.
- **Q2 (median)**: The middle value of the data set, which divides the data into two equal halves.
- **Q3 (75th percentile)**: The value that separates the lowest 75% of the data. 75% of the data points are less than Q3.

IQR: the difference between the 75th percentile (Q3) and the 25th percentile (Q1)

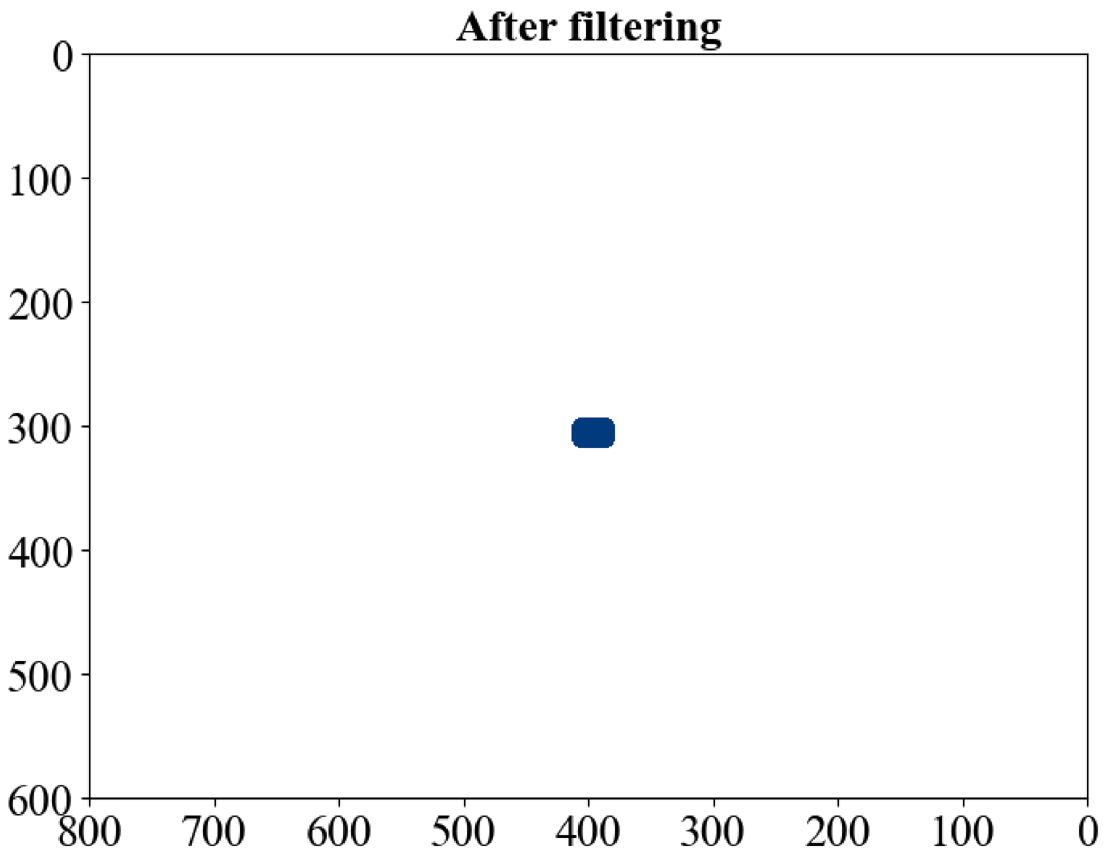


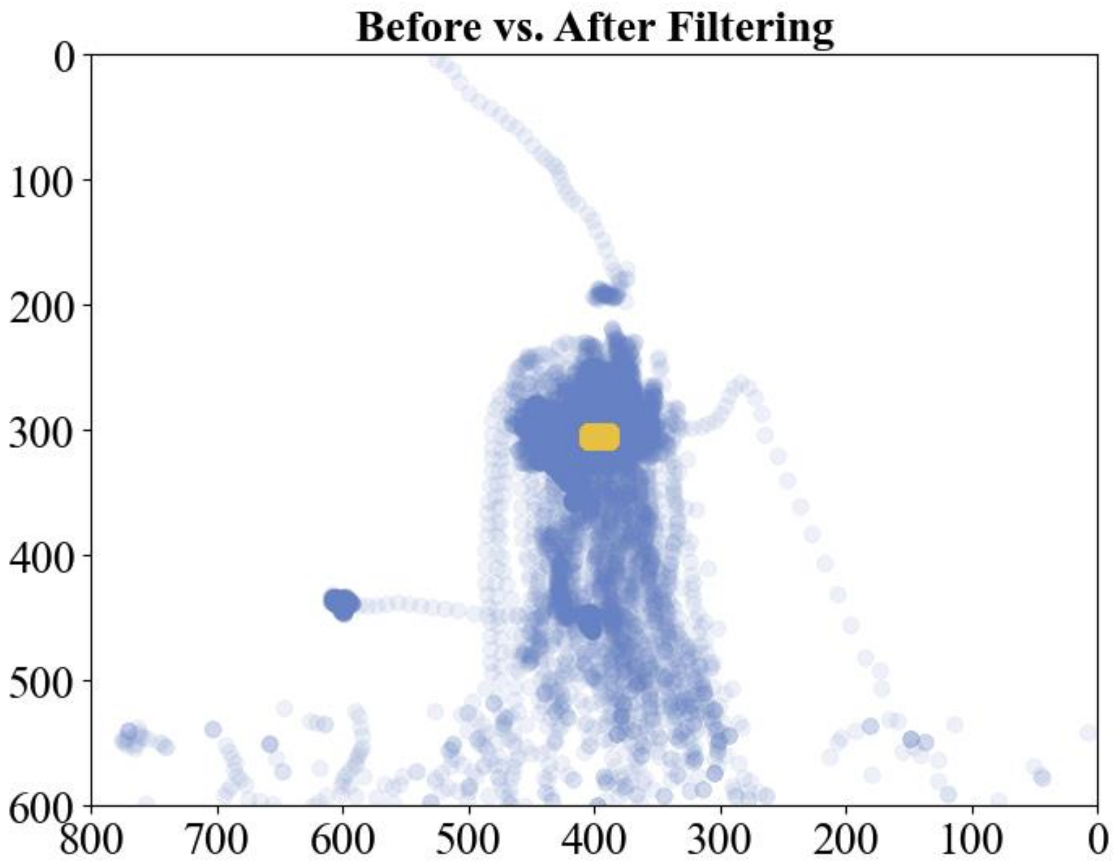


Upper boundary = Q3
Lower boundary = Q1
Small IQR -> The majority of gaze points are close to the median

Use case 2: Eye-Tracker

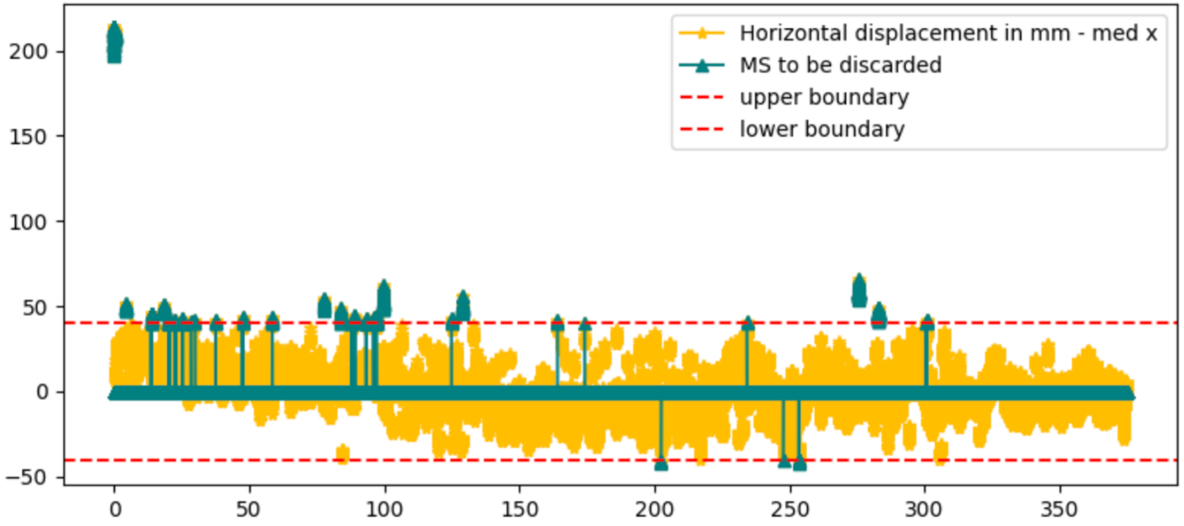
Gaze points located between Q1 and Q3

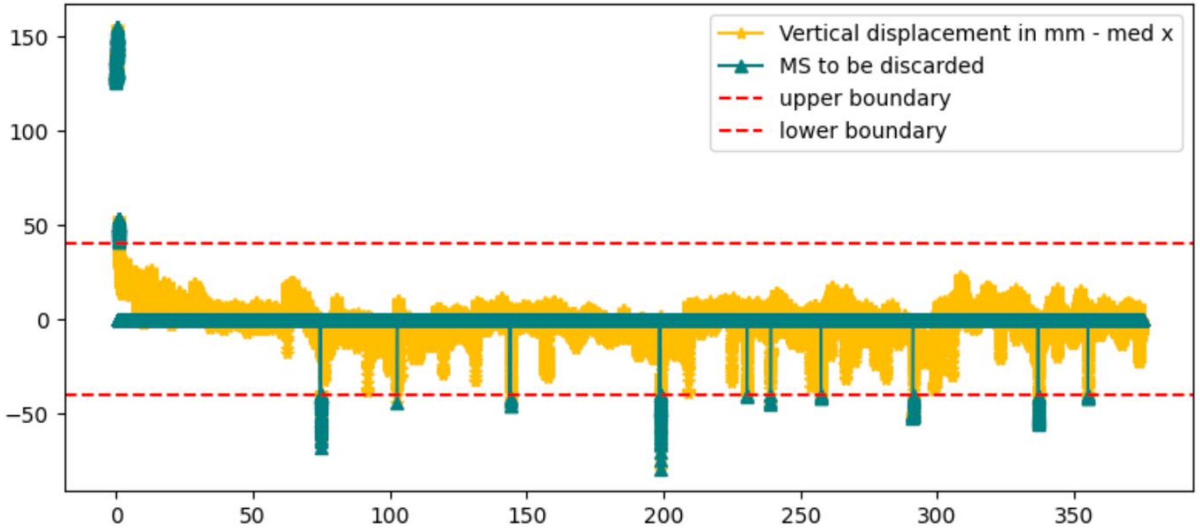




Use case 2: Eye-Tracker

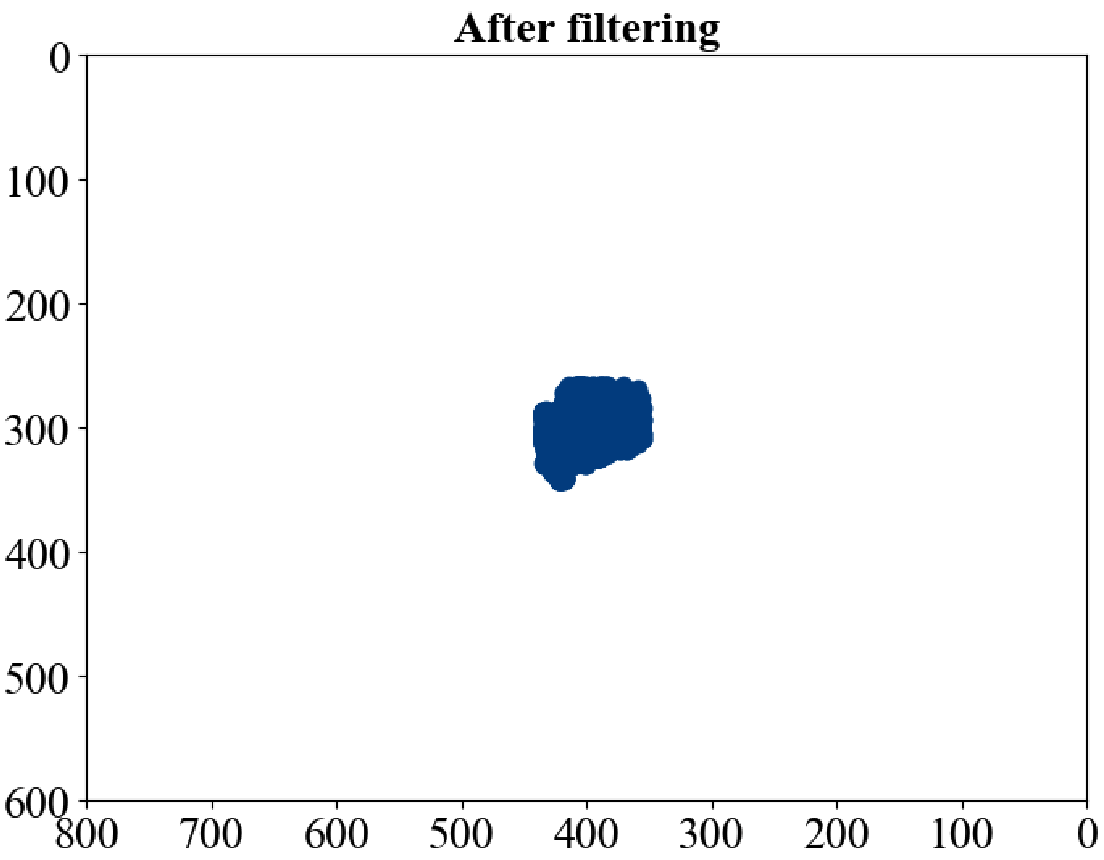
Adjust different upper and lower boundaries according to specific needs

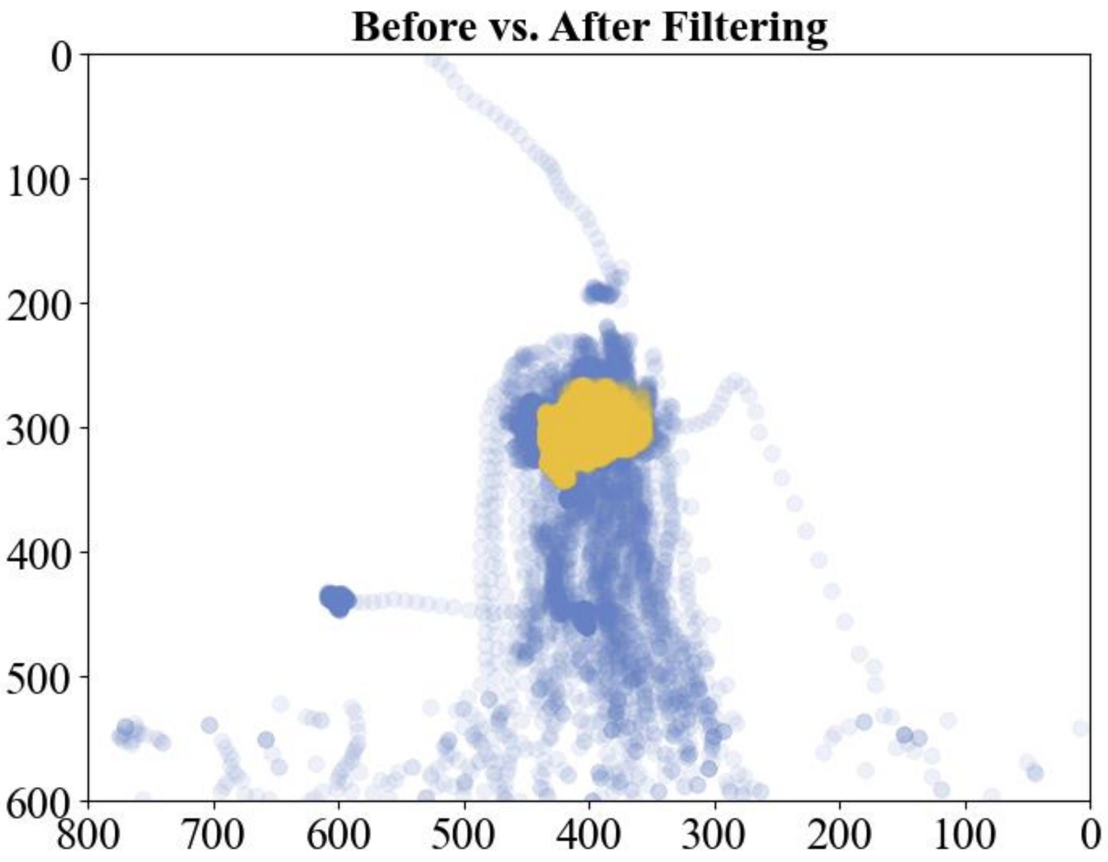




Use case 2: Eye-Tracker

Corresponding filtering results





Use case 2: Eye-Tracker

Investigate the eye movement features across 4 subjects

Blink (ms)

Subject	Mean Duration	Median Duration	Standard Deviation	Min Duration	Max Duration	Total Duration	Number of Times
1	307.87	298.0	102.14	136	612	9544	31
2	193.79	191.0	53.48	84	366	8333	43
3	279.44	257.0	95.56	82	643	17605	63
4	570.57	222.5	963.50	2	7006	94714	166

Saccade (ms)

Subject	Mean Duration	Median Duration	Standard Deviation	Min Duration	Max Duration	Total Duration	Number of Times
1	30.50	17.0	63.47	6	612	22203	728
2	36.31	22.0	50.18	6	366	19206	529
3	51.54	26.0	79.13	6	643	36796	714
4	151.31	63.0	390.59	2	7006	202600	1339

Subject 004
looks different?

blinks and saccade
- Long total duration
- Occurred many times

Use case 2: Eye-Tracker

Investigate the eye movement features across 4 subjects

Fixation

Subject	Mean Duration	Median Duration	Standard Deviation	Min Duration	Max Duration	Total Duration	Number of Times
1	485.33	247.0	624.00	22	4157	353804	729
2	672.02	491.5	594.36	17	5003	356169	530
3	417.98	246.0	466.94	24	3348	298856	715
4	128.82	92.0	117.28	9	1012	172486	1339

Subject 004
looks different?

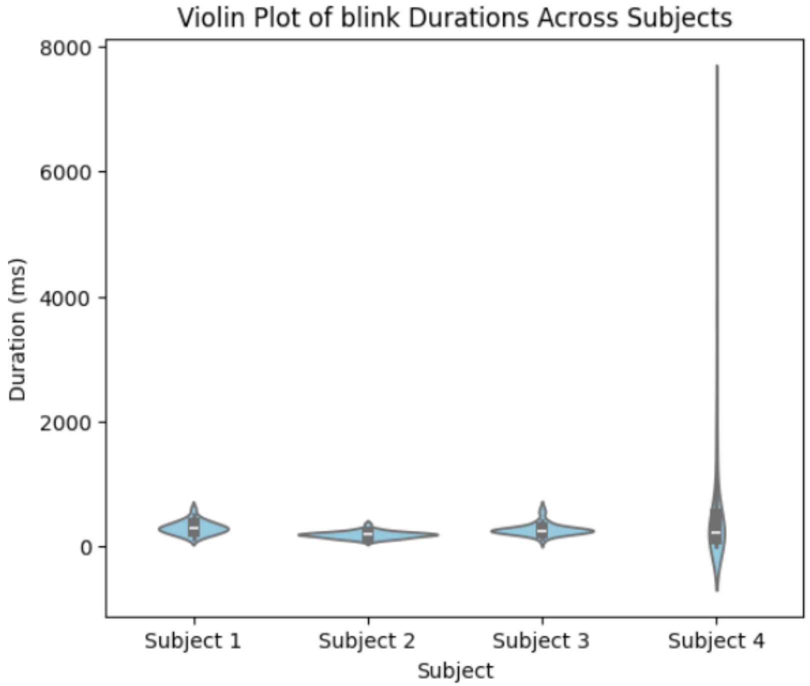
Fixation
- → Short duration

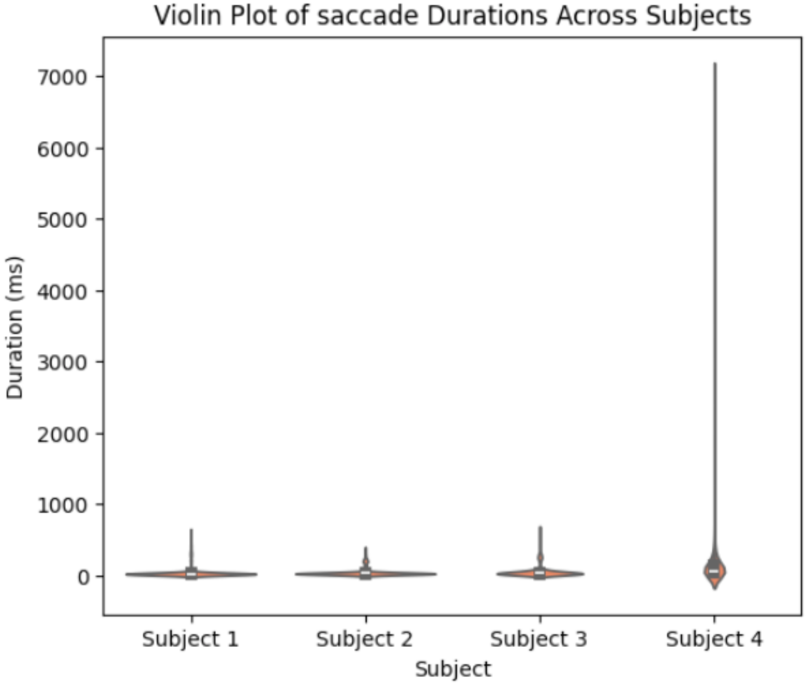
Reflect the data quality
- ✓ Think about quality control

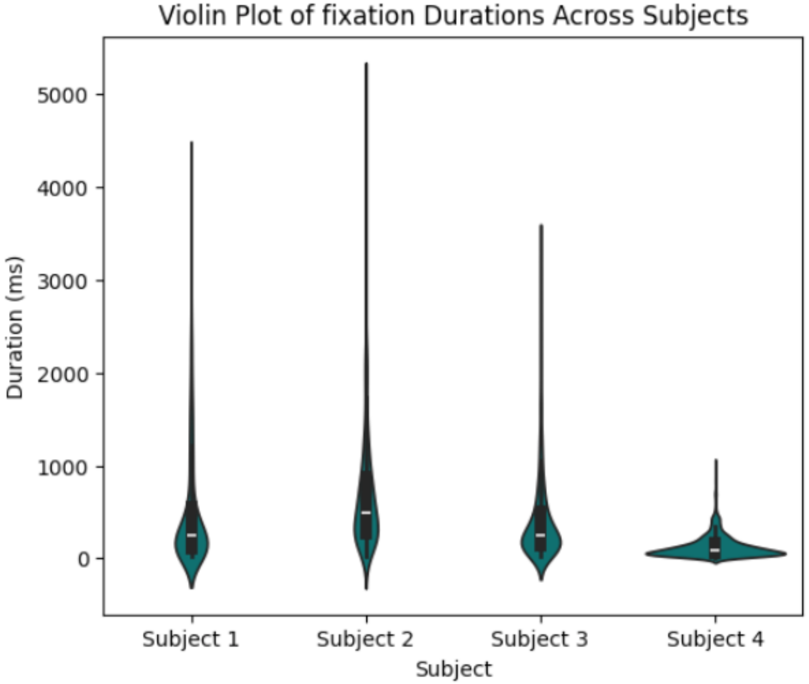
Use case 2: Eye-Tracker

Visualize the statistics of event durations across four subjects.

Violin plots are handy tools







Think about quality control of early data:

- Poor eye tracker calibration?
- Subject was tired?
- ...

Is good for

- ✓ Retrospectively update the SOP of your experiment.
- ✓ Predict results in future steps.
- ✓ Stop the processing loop and save more time

Hand-on session: Eye-Tracker

https://github.com/MattechLab/SigInf_2025

https://colab.research.google.com/